

Harrison Crosse

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EMPLOYMENT

Docker

Senior Software Engineer 2025 - Present

- Co-designed Marlin, Docker's event platform for tens of thousands of events per second. Established its SDK and ingestion architecture and guided expansion into stream processing and audit-log analytics.
- Built Marlin SDKs for browser, Node.js, and Rust around shared protobuf contracts. Productionized them with batching, retries, integration tests, and release automation.
- Designed Marlin's privacy architecture and annotation-driven PII handling across SDKs, ingestion workers, and Iceberg, supporting Docker's ISO 27701 certification.
- Hardened Marlin ingestion with schema validation, version-gated migrations, DLQs, and production observability.

Calendly

Senior Data Engineer 2024 - 2025

Data Engineer 2022 - 2024

- Built Calendly's data platform from the ground up, migrating Postgres and event data from Redshift to BigQuery and establishing Airflow, CI/CD, data quality, and developer tooling.
- Processed 50 TB of malformed Segment replay data with Spark for BigQuery.
- Built production Flink pipelines processing thousands of records per second with hundreds of gigabytes of state. Added observability, blue-green deployments, data-quality checks for cutovers, and E2E tests.
- Replaced unreliable Google Sheets-defined schemas with a GitOps Data Contracts workflow across hundreds of Segment event types.

Amobee

Software Engineer, Data Systems 2021 - 2022

- Moved clickstream processing from batch Spark to Structured Streaming, making data continuously available downstream.
- Moved Spark workloads from EMR to Kubernetes with Spark Operator, standardizing deployments and eliminating EMR service charges.

Sayari Labs

Data Engineer 2020 - 2021

- Built scraping and fastText NER pipelines to transform corporate registry data into structured entities and social graphs.
- Migrated Sayari's cloud infrastructure from GCP to AWS GovCloud using Terraform and Docker.

Booz Allen Hamilton

Data Scientist 2019 - 2020

- Modeled Marine Corps unit performance for the Inspector General using clustering and gradient-boosted trees.

EDUCATION

University of Mary Washington

B.S. Computer Science 2019

Minor in Data Science

SELECTED WORK

DataFusion / Ballista

Upstreamed non-blocking distributed shuffle execution by exposing async batch partitioning in DataFusion and offloading Ballista shuffle writes from query workers.

DataFusion #21341 Ballista #1537

Apache Iceberg / Arrow Go

Improved Parquet writer compatibility by adding table-level page-version configuration to Iceberg and fixing invalid root-schema serialization in Arrow Go.

Iceberg #15700 Arrow Go #723

TOOLS

LANGUAGES

[Bash](#) [C](#) [Go](#) [Java](#) [Python](#)
[Rust](#) [SQL](#) [TypeScript](#)

DATA SYSTEMS

[Airflow](#) [Arrow](#) [BigQuery](#)
[ClickHouse](#) [dbt](#) [Debezium](#)
[Flink](#) [Iceberg](#) [Kafka](#)
[Parquet](#) [PostgreSQL](#) [Protobuf](#)
[Snowflake](#) [Spark](#)

INFRASTRUCTURE

[Argo](#) [AWS](#) [Docker](#) [GCP](#)
[GitHub Actions](#) [Grafana](#) [Helm](#)
[Kubernetes](#) [Linux](#) [Terraform](#)